W-C-583b
March 10, 1965
SUPERSEDING
Int. Fed. Spec. W-C-00583a
July 8, 1963 and
Fed. Spec. W-C-583
July 31, 1957

FEDERAL SPECIFICATION

CONDUIT BOXES AND OUTLET FITTINGS, FLOOR, (FOR RIGID METAL CONDUIT)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers watertight and concrete-tight, cast-metal and pressed steel floor conduit boxes for use with rigid metal conduit (thickwall-type and electrical metallic tubing); and the outlet fittings for use with these boxes.

1.2 Classification.

1.2.1 *Floor conduit boxes.*

1.2.1.1 *Types, classes, and styles.* Floor conduit boxes furnished under this specification shall be of the following types, classes, and styles, as specified (see 6.1).

Types:

I—Adjustable
II—Semiadjustable
III—Nonadjustable

Classes:

1—Watertight
2—Concrete-tight

Styles:

A—Round
B—Rectangular

1.2.2 *Floor outlet fittings.*

1.2.2.1 *Types.* Floor outlet fittings furnished under this specification shall be of the type specified (see 3.3.9 and 6.1).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-585—Boxes, Wood, Wirebound.
PPP-B-591—Boxes, Fiberboard, Wood-Cleated.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-B-676—Boxes, Set-Up Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S.

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Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-10547—Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.
MIL-P-116—Preservation, Methods of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.
MIL-STD-1186—Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with Appropriate Test Methods.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following publications, of the issues in effect on date of invitation for bids, form a part of this specification.

Underwriter's Laboratories, Inc., (UL) Standard:

UL-514—Outlet Boxes and Fittings.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., Walt Whitman Road, New York, N. Y., 11749; 207 East Ohio Street, Chicago, Illinois, 60611; and 1655 Scott Blvd., Santa Clara, California, 95050.)

National Bureau of Standards Handbook:

H28—Screw-Thread Standards for Federal Services.

(Copies may be obtained upon application accompanied by money order, coupon, or cash to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Price may be obtained from the Superintendent of Documents.)

3. REQUIREMENTS

3.1 **Fire and casualty hazards.**

3.1.1 Each contractor shall submit to the contracting agency proof that the material he proposes to supply under this specification conform to the requirements of the Underwriters' Laboratories, Inc., Standard UL-514 for Outlet Boxes and Fittings. The label, or listing with re-examination, of the Underwriters' Laboratories, Inc., may be accepted as evidence that the material conforms to this requirement.

3.1.2 In lieu of the label, or listing with re-examination, the contractor may submit independent proof satisfactory to the contracting agency, that the equipment conforms to the applicable requirements of the published standards including methods of tests of UL-514.

3.1.3 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure acceptance of his material.

3.2 Except as otherwise provided herein, the floor conduit boxes and outlet fittings covered by this specification shall conform to the applicable requirements of UL-514.

3.3 **Detail requirements.**

3.3.1 *Type I conduit boxes.* Adjustable floor conduit boxes shall be constructed so that the cover plate may be adjusted vertically not less than 3/8-inch from the plane of its base, and shall have an angular leveling adjustment of not less than 10 degrees for use after box body is secured permanently.

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3.3.2 *Type II conduit boxes.* Semiadjustable floor conduit boxes shall be constructed so that the cover plate may be adjusted vertically not less than 3/8-inch from the plane of its base, and shall have an angular leveling adjustment of not less than 10 degrees for use at time of installation.

3.3.3 *Type III conduit boxes.* Nonadjustable floor conduit boxes shall be constructed without means for angular or vertical adjustment of the cover plate.

3.3.4 *Class 1 conduit boxes.* Watertight floor conduit boxes shall be constructed so as to prevent the entrance of both surface and sub-surface water into the outlet in all locations. They shall be provided with integral threaded conduit opening of sizes as required (see 6.1), having at least four clean, sharp, and full depth American National straight or taper pipe threads. All threaded parts shall comply with Handbook H28.

3.3.5 *Class 2 conduit boxes.* Concrete-tight floor conduit boxes shall be constructed so as to prevent the free entrance of both surface and sub-surface water when the box body is embedded in concrete in accordance with UL-514. They shall have knockouts of sizes as required (see 6.1).

3.3.6 *Gang floor outlets.* Gang floor outlets shall be provided with partitions between each section in the body and in the adjusting flange, if of the adjustable type.

3.3.7 *Mounting provisions.* When floor outlet bodies are provided with mounting means for securing the outlet in position, such mounting means shall be external. Provisions for support of devices to be enclosed in outlets shall be provided with an effective electrical bond for grounding the cover plate to the body.

3.3.8 *Cover plates.* Cover plates shall have a minimum thickness of 0.125 inch.

3.3.8.1 Cover plate gaskets shall be supplied, and shall be made of cloth-inserted rubber not less than one-sixteenth inch thick, or equivalent semi-soft resilient material.

3.3.8.2 Leveling and adjusting screws shall be made of brass or bronze or of steel. Steel screws shall be protected against corrosion with cadmium or zinc coatings.

3.3.8.3 Cover-securing screws, if exposed after installation, shall be made of brass or bronze and shall be not smaller than No. 6.

3.3.9 *Floor outlet fittings.* Floor outlet fittings shall include straight multiple, and bell nozzles, standpipes, outlet and similar fittings designed for attachment to the floor outlet. The interior surfaces of floor outlet fittings, with which the conductors may come in contact, shall be smooth and free from projections which might cause abrasion of the insulation on the conductors. All metal parts of floor outlet fittings shall be made of brass, bronze, aluminum, or other metals suitable in strength and noncorrosive properties.

3.3.10 Exposed parts of floor conduit boxes and floor outlet fittings shall be made of brass, bronze, aluminum, or other metals suitable in strength and noncorrosive properties.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure that supplies and services conform to prescribed requirements.

4.2 Inspection lot. All floor conduit boxes or fittings of the same type, class and size presented at one time shall be considered as a lot for the purpose of acceptance, inspection, and tests.

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4.3 **Sampling.**

4.3.1 *Visual and dimensional inspection.* Samples shall be selected from each inspection lot in accordance with inspection level II of MIL-STD-105 and examined by the Government inspector to verify compliance with this specification for all characteristics that may be determined by visual and dimensional inspection. Acceptance shall be based on an acceptable quality level (AQL) of 2.5. The Government inspector may institute tightened or reduced inspection in accordance with paragraph 8 of MIL-STD-105.

4.3.2 *Lot acceptance tests.* Samples shall be selected at random from each inspection lot by the Government inspector for lot acceptance tests in accordance with inspection level S-3 of MIL-STD-105. Acceptance shall be based on an AQL of 6.5.

4.4 **Lot acceptance test.** Tests to determine compliance with this specification shall be in accordance with the test procedures specified in UL-514 for outlet boxes and fittings.

4.5 **Acceptance inspection of preparation for delivery.** An inspection shall be made to determine compliance with packaging, packing, and marking requirements of section 5.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and application of levels of preservation, packaging, and packing shall be in accordance with Fed. Std. No. 102.)

5.1 **Preservation and packaging.**

5.1.1 *Level A.*

5.1.1.1 *Floor conduit boxes.* Floor conduit boxes shall be preserved and packaged in accordance with MIL-P-116, method III.

5.1.1.2 *Fittings.* The fittings shall be packaged in paperboard folding boxes conforming to PPP-B-566, or paperboard set-up boxes conforming to PPP-B-676. The gross weight of the boxes shall not exceed the weight limitations of the applicable box specification.

5.1.2 *Level C.* The floor conduit boxes and fittings shall be packaged in accordance with the standard commercial practice.

5.2 **Packing.**

5.2.1 *Level A.* The floor conduit boxes and fittings shall be packed in nailed wood, wood-cleated-plywood, wirebound wood, wood-cleated-fiberboard, or fiberboard boxes, conforming to PPP-B-621, class 2; PPP-B-601 (overseas type); PPP-B-585, class III; PPP-B-591; and PPP-B-636, compliance symbol V3s or V3c, respectively. Boxes shall be lined with waterproof case liners conforming to MIL-L-10547, sealed in accordance with the appendix thereto. Case liners for boxes conforming to PPP-B-636, are not required provided all corners and joints of the box are waterproof sealed with tape in accordance with the appendix to the fiberboard box specification. Contents of boxes shall be cushioned and blocked within the boxes in accordance with MIL-STD-1186. Boxes shall be closed and strapped in conformance with the applicable box specification or the appendix thereto, except straps shall be flat steel, type I, class B, grade 2. The gross weight of wood boxes shall not exceed approximately 200 pounds and of fiberboard boxes 70 pounds.

5.2.2 *Level B.* The floor conduit boxes and fittings shall be packed in nailed wood, wood-cleated-plywood, wirebound wood, wood-cleated-fiberboard, corrugated fiber or solid fiber boxes conforming to PPP-B-621, class 1; PPP-B-601 (domestic type); PPP-B-585, class 1; PPP-B-591; PPP-B-636, respectively. Cushioning and blocking shall be provided to prevent movement within the boxes. The gross weight of each wood or wood-cleated fiberboard box shall not exceed 200 pounds.

5.2.3 *Level C.* The floor conduit boxes and fittings shall be packed to insure carrier acceptance and safe delivery to destination in

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compliance with carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with Fed Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required in the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.
(*b*) Type, class, and style of floor conduit box (1.2.1.1).
(*c*) Type of outlet fittings (1.2.2.1 and 3.3.9).
(*d*) Conduit sizes to be accommodated (3.3.4 and 3.3.5).
(*e*) Number of gangs (style B outlets only).
(*f*) Responsibility for inspection, if other than specified (4.1).
(*g*) Level of packaging and packing required (5.1 and 5.2).
(*h*) Special marking, if required (5.3).

6.2 Transportation description. Transportation descriptions and minimum weights applicable to this commodity:

Rail:

Conduit outlets and fittings, brass.
Carload minimum weight 30,000 pounds.

Motor:

Conduit outlets and fittings, brass.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—MO (ERDL)

Navy—YD

Air Force—85

Review Interest:

Army—MO (ERDL)

Navy—YD

Air Force—85

User Interest:

Army—MU

Navy—MC

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